IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies-

1. Data Collection

2. Data Wrangling

3. Exploratory Data Analysis (EDA) using visualization and SQL

- Used **Matplotlib, Seaborn, and Plotly** for visual exploration.
- Analyzed payload mass vs. landing success.
- Compared launch sites and booster versions to identify success patterns.
- **4. Feature Engineering**
- Created derived features (e.g., binary success indicator).
- Applied normalization and scaling to improve model performance.

5. Machine Learning Models

- Implemented classification models: **Logistic Regression, Decision Tree, KNN, SVM**.
- Evaluated models using **train/test split** and **cross-validation**.
- Measured performance with **accuracy, precision, recall, and F1-score**.

6. Model Comparison

- Compared all models to identify the best predictor of Falcon 9 landing success.
- Highlighted trade-offs between interpretability and accuracy.

7. Insights & Conclusion

- Landing success strongly depends on **launch site, payload mass, and orbit type**.
- Machine learning models can reliably predict Falcon 9 landing outcomes.
- Findings validate SpaceX's reusable rocket strategy as a cost-saving innovation.

Summary of all results

🚀 **Launch Success Trends**

- **Success rates improved over time**: Higher flight numbers show more consistent success, especially for LEO missions.
- **Yearly trend**: Average success rate increased steadily, indicating operational maturity.

🛰️ **Orbit Type Analysis**

- **Highest success**: Orbits like ES-L1, GEO, HEO, SSO, VLEO had near-perfect success rates.
- **Moderate success**: LEO, MEO, PO, GTO showed variable success, with GTO slightly lower.
- **Lowest success**: ISS orbit had mixed outcomes, suggesting mission complexity.

📍 **Launch Site Insights**

- **KSC LC 39A**: Most reliable for heavy payloads and high flight numbers.
- **VAFB SLC 4E**: Specialized for lighter payloads; high success rate.
- **CCAFS SLC 40**: Mixed results across payloads and flight numbers.

📦 **Payload Mass Patterns**

- **Heavy payloads**: More successful in Polar, LEO, and ISS orbits.
- **GTO missions**: No clear correlation between payload mass and success.

🔍 **Feature Engineering**

- Created dummy variables for categorical features (Orbit, LaunchSite, LandingPad, Serial).
- Cast all numeric features to float64 for ML readiness.
- Final dataset exported as dataset_part_3.csv.

Introduction

Project Background – Winning Space Race with Data Science

The commercial space age is accelerating, with companies like **Virgin Galactic**, **Rocket Lab**, **Blue Origin**, and **SpaceX** making space travel more accessible. Among them, **SpaceX** stands out for its cost-effective launches—thanks to its ability to **reuse the Falcon 9 first stage**.

A single Falcon 9 launch costs **$62 million**, compared to **$165 million** from other providers. Much of this saving comes from **recovering the first stage**, which is large, expensive, and mission-critical.

However, **not all missions allow recovery**—some stages crash or are sacrificed due to payload, orbit, or customer demands.

In this Space Race, I As a **data scientist at Space Y**, a fictional competitor to SpaceX founded by billionaire **Allon Musk**.

Mission:

Predict whether Falcon 9's first stage will land successfully.

Estimate launch costs using public SpaceX data.

Build dashboards and train machine learning models to support strategic decisions.

Section 1

Methodology

Methodology

Executive Summary

Data collection methodology:

- Retrieved launch data from the **SpaceX API**.
- Supplemented with additional information via **web scraping**.
- Consolidated datasets using **Pandas** for structured analysis.

Perform data wrangling

- Cleaned missing values and standardized inconsistent entries.
- Selected key features: payload mass, launch site, orbit, booster version, landing outcome.
- Converted categorical variables into numerical form for modeling.

Perform exploratory data analysis (EDA) using visualization and SQL

Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using classification models

- Implemented classification models: **Logistic Regression, Decision Tree, KNN, SVM**.
- Evaluated models using **train/test split** and **cross-validation**.
- Measured performance with **accuracy, precision, recall, and F1-score**.

Data Collection

- Retrieved launch data from the **SpaceX API**.
- Supplemented with additional information via **web scraping**.
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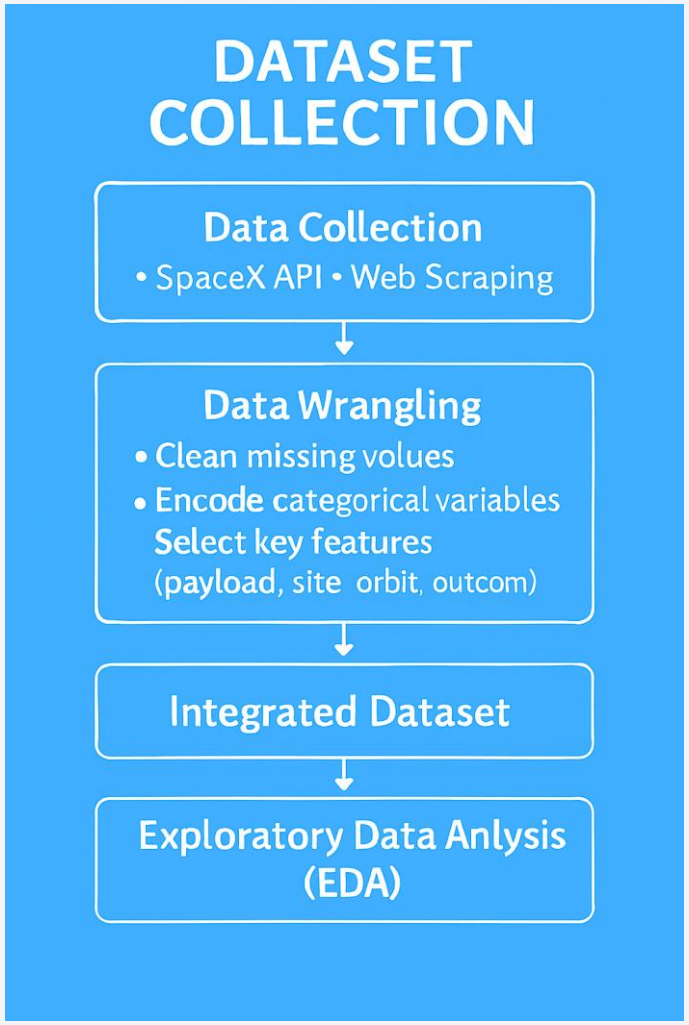
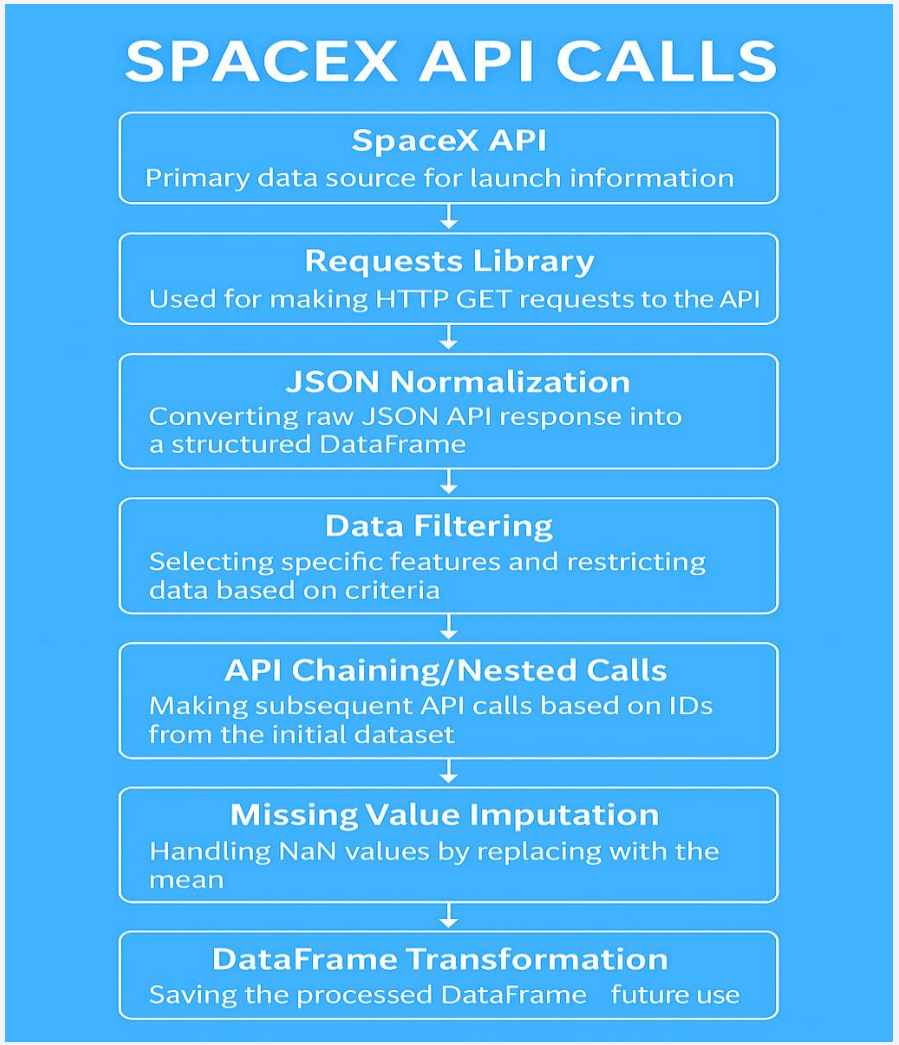


Data Collection – SpaceX API

- Data collection with SpaceX REST calls using key phrases and flowcharts

- The GitHub URL of the completed SpaceX API calls notebook

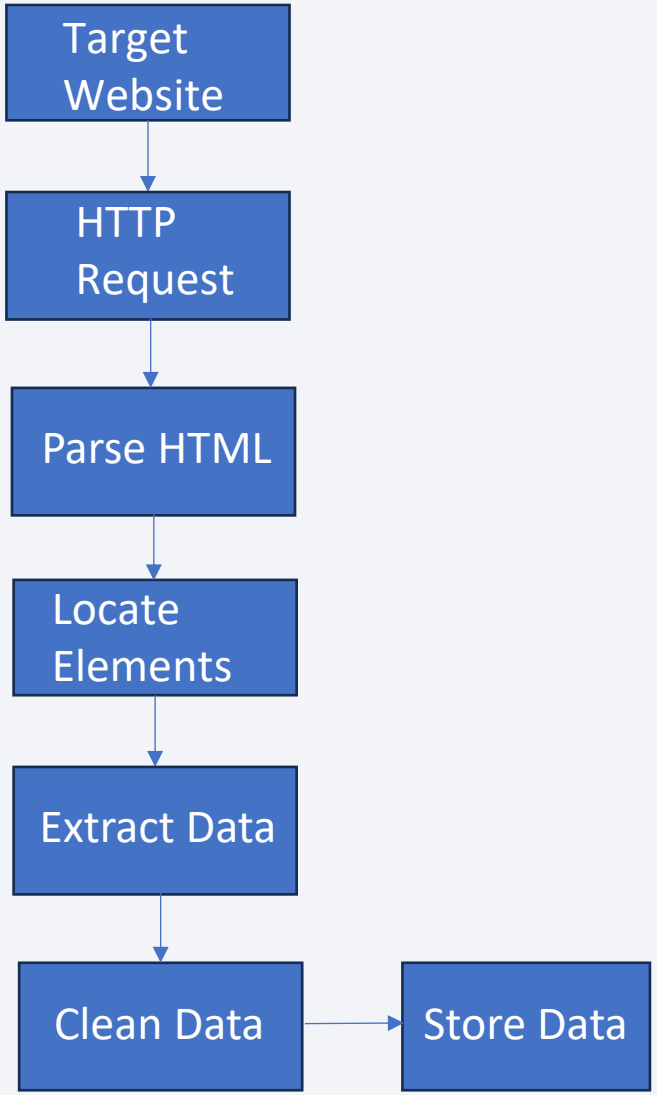
- https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/SpaceXAPIDataCollection.ipynb



Data Collection - Scraping

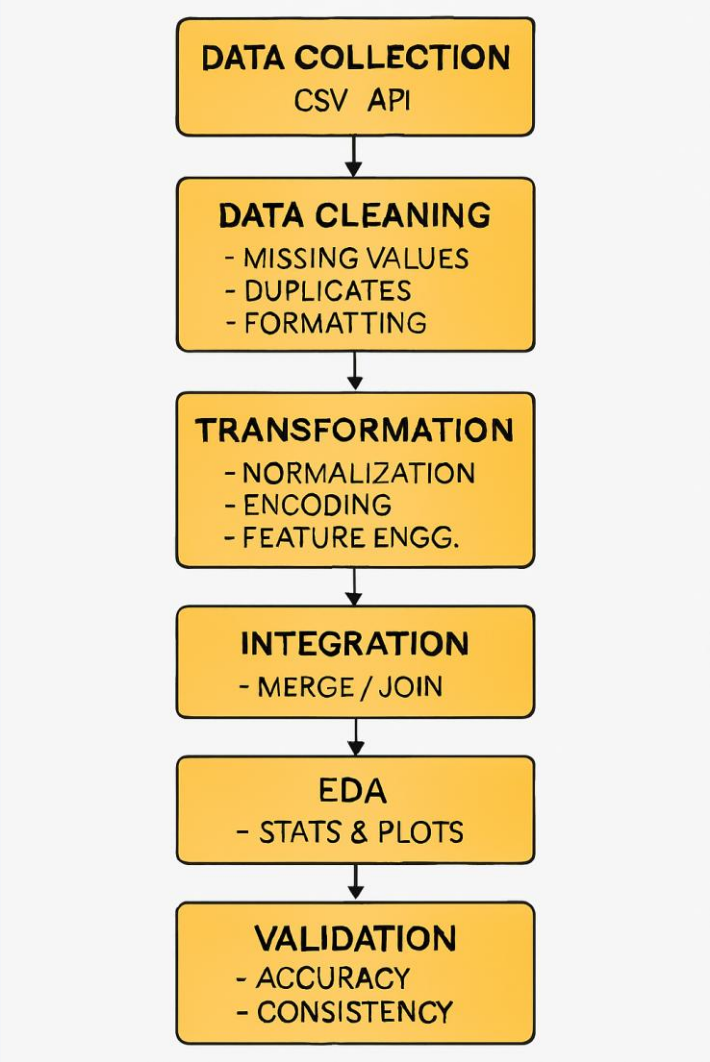
Key Phrases for Web Scraping Workflow

- Target Website → Define the source of data"https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"
- HTTP Request → Fetch HTML using requests
- Parse HTML → Use BeautifulSoup or lxml
- Locate Elements → Identify tags, attributes, CSS selectors
- Extract Data → Pull text, links, tables
- Clean Data → Format, remove duplicates, handle missing values
- Store Data → Save to CSV, JSON, or database
- Automate → Loop through pages, handle dynamic content
- **Github URL**
- [WinningSpaceRaceWithDataScience/jupyter-labs-webscraping.ipynb at main · Sumant-crty/WinningSpaceRaceWithDataScience]



Data Wrangling

- GitHub URL
- https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

- **1. Scatter Plot (Payload Mass vs. Launch Outcome)**
- **Why:** To examine the relationship between payload mass and launch success.
- Helps identify whether heavier payloads affect the probability of success.
- **2.Scatter Plot(Flight Number and Launch Site)**
- Why: To examine ,how launches are distributed across different sites over time (or by sequence).
- **Helps to identify that Cape Canaveral (CCAFS)**: Appears most often, indicating it is the primary site for SpaceX launches.
- **2. Bar Chart (Launch Site vs. Success Count)**
- **Why:** To compare the number of successful launches across different launch sites.
- Highlights which sites have higher success rates

GitHub URL of our completed EDA with data visualization notebook, as an external reference and peer-review purpose

https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/edadataviz.ipynb

EDA with SQL

The SQL queries performed

- create table SPACEXTABLE as select * from SPACEXTBL where Date is not null
- **SELECT DISTINCT** Launch_Site **FROM** SPACEXTABLE;
- **SELECT * FROM** SPACEXTABLE **WHERE** Launch_Site **LIKE** 'CCA%' **LIMIT** 5;
- **SELECT SUM**(Payload_Mass__kg_) **FROM** SPACEXTABLE **WHERE** Customer **=** 'NASA (CRS)';
- **SELECT AVG**(Payload_Mass__kg_) **FROM** SPACEXTABLE **WHERE** Booster_Version **LIKE** 'F9 v1.1%';
- **SELECT MIN**(Date) **FROM** SPACEXTABLE **WHERE** Landing_Outcome **=** 'Success (ground pad)';
- **SELECT DISTINCT** Booster_Version **FROM** SPACEXTABLE **WHERE** Landing_Outcome **=** 'Success (drone ship)' **AND** Payload_Mass__kg_ **>** 4000 **AND** Payload_Mass__kg_ **<** 6000;
- **SELECT** Mission_Outcome, **COUNT**(Mission_Outcome) **AS** Total **FROM** SPACEXTABLE **GROUP BY** Mission_Outcome;
- **SELECT DISTINCT** Booster_Version **FROM** SPACEXTABLE **WHERE** Payload_Mass__kg_ **=** (**SELECT MAX**(Payload_Mass__kg_) **FROM** SPACEXTABLE);
- **SELECT** substr(Date, 6, 2) **AS Month**, Booster_Version, Launch_Site **FROM** SPACEXTABLE **WHERE** substr(Date, 0, 5) **=** '2015' **AND** Landing_Outcome **=** 'Failure (drone ship)';
- **SELECT** Landing_Outcome, **COUNT**(Landing_Outcome) **AS** Total_Count **FROM** SPACEXTABLE **WHERE** Date **BETWEEN** '2010-06-04' **AND** '2017-03-20' **GROUP BY** Landing_Outcome **ORDER BY** Total_Count **DESC**;

GitHub URL

https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map objects created and added to a folium map and use to represent-

- **Circle**: You add circles to represent points of interest with a specific radius. This is useful for showing areas of influence, proximity zones, or the approximate location of something that isn't a precise point, like the range of a signal or a general region around a launch site.
- **Marker**: Markers are used for precise point locations on the map. They are ideal for indicating exact spots like specific launch pad locations, individual buildings, or single points of data. Markers can often have popups with additional information.
- **PolyLine**: PolyLines are used to draw lines or paths on the map. They are essential for visualizing routes, boundaries, connections between two points (like the distance lines we drew from the launch site to the coastline), or any linear geographical feature.
- **MarkerCluster**: This is a plugin used to manage a large number of markers. When many markers are close together, they can clutter the map and make it unreadable.
  A MarkerCluster automatically groups nearby markers into a single cluster icon. As you zoom in, the cluster expands to reveal individual markers. This improves map performance and user experience by presenting dense data in an organized way.

- **GitHub URL of completed interactive map with Folium map,**
- https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Two main plots added to this dashboard:
- **Success Pie Chart**: This chart dynamically displays launch success rates. If 'All Sites' is selected, it shows the total successful launches by each site. If a specific launch site is chosen, it illustrates the proportion of successful versus failed launches for that particular site.
- **Success Payload Scatter Chart**: This scatter plot visualizes the correlation between the payload mass (in kg) and the launch outcome (success or failure). It allows users to filter by launch site and payload range, and categorizes booster versions by color, providing insights into how payload mass and booster type affect launch success.

GitHub URL of your completed Plotly Dash lab

https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/Plotlyexample.ipynb

Predictive Analysis (Classification)

A comprehensive summary of the entire process, including the steps taken to build, evaluate, and identify the best classification model, highlighting key findings for each model and the overall best performer.

- **Data Preparation**: Outline the steps taken for data loading, target variable creation (Y), feature standardization (X), and splitting the data into training and testing sets.
- **Model Building and Tuning**: Describe the construction of each classification model (Logistic Regression, SVM, Decision Tree, K-Nearest Neighbors) and how GridSearchCV with cross-validation (cv=10) was used to tune their hyperparameters and find the best parameters for each.
- **Model Evaluation**: Explain how each model's performance was evaluated using GridSearchCV's best_score_ for cross-validation accuracy and the .score() method on the test data (X_test, Y_test) for final accuracy. Also, mention the use of confusion matrices.
- **Identify Best Performing Model**: Compare the test accuracies of all trained models (Logistic Regression, SVM, Decision Tree, K-Nearest Neighbors) and identify which model achieved the highest accuracy on the unseen test data, along with its specific accuracy score.
- **Github URL**

https://github.com/Sumant-crty/WinningSpaceRaceWithDataScience/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Prediction Result

Compared four different classification models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). Here are their performance metrics:

Logistic Regression

Cross-Validation Score: 0.8464
Test Accuracy: 0.8333

Support Vector Machine (SVM)

Cross-Validation Score: 0.8482
Test Accuracy: 0.8333

Decision Tree

Cross-Validation Score: 0.8875
Test Accuracy: 0.8333

K-Nearest Neighbors (KNN)

Cross-Validation Score: 0.8482
Test Accuracy: 0.8333

Based on these results, the **Decision Tree** model performed best with the highest cross-validation score of 0.8875, although all models achieved the same test accuracy of 0.8333.

Based on the Exploratory Data Analysis performed in the notebook, here are the key results:

Missing Values: The LandingPad column has approximately 28.89% missing values, while all other columns have no missing values.

Data Types: Columns like Date, BoosterVersion, Orbit, LaunchSite, Outcome, and Serial are of object (string) type, PayloadMass, Block, Longitude, Latitude are floats, FlightNumber, Flights, ReusedCount are integers, and GridFins, Reused, Legs are boolean.

Launches per Site:

CCAFS SLC 40: 55 launches
KSC LC 39A: 22 launches
VAFB SLC 4E: 13 launches

Launches per Orbit (excluding GTO): (The output for this was not displayed in the provided state, but the code to calculate it was present. I will mention this and ask the user if they want me to display the output explicitly.)

Mission Outcomes:

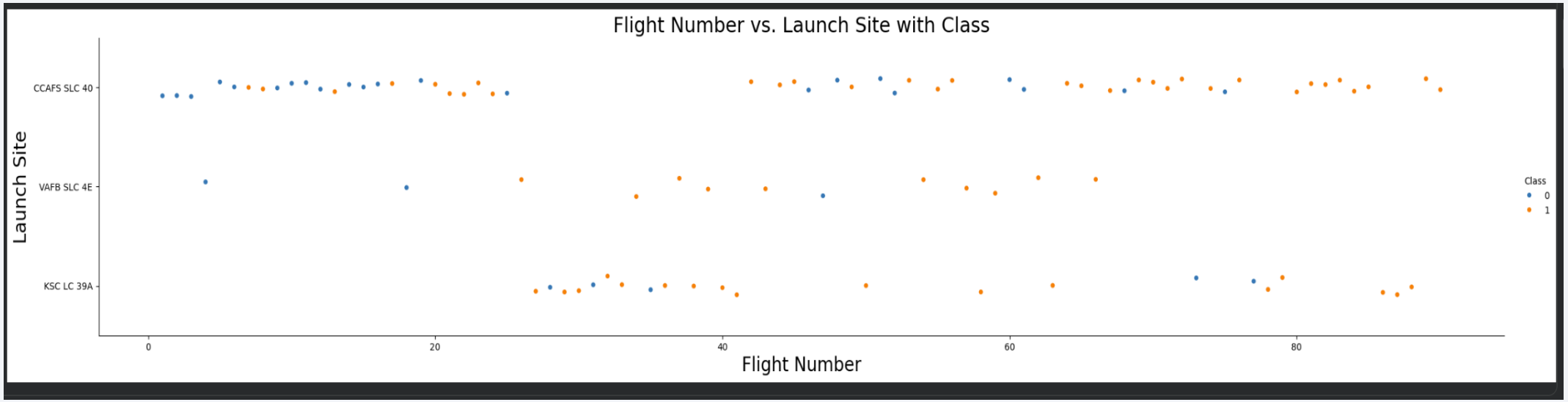
True ASDS: 41
None None: 19
True RTLS: 14
False ASDS: 6
True Ocean: 5
False Ocean: 2
None ASDS: 1
False RTLS: 1

Success Rate: The overall success rate for the first stage landing is approximately 66.67%.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



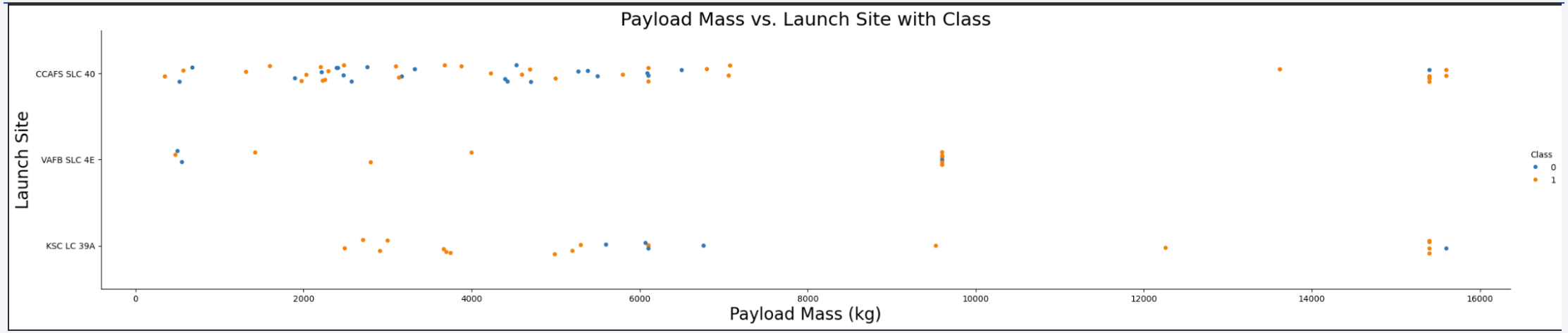
CCAFS SLC 40: This launch site shows a high concentration of early flights, particularly those with a 'Class' of 0 (unsuccessful landing). As the flight number increases, there's a clear trend towards more successful landings (Class 1) at this site.

VAFB SLC 4E: This site has fewer launches overall. For early flight numbers, success rates appear mixed, but with increasing flight numbers, there's a tendency towards successful landings.

KSC LC 39A: This site seems to have been introduced later in the sequence of flight numbers. It generally shows a very high success rate from its inception, with most launches resulting in successful landings.

In general, across all sites, there's a visible trend: as the flight number increases, the probability of a successful landing (Class 1) also increases, indicating improved technology, experience, and reliability over time. Different launch sites also appear to have different success rates, possibly due to factors like their operational history and the type of missions they typically host.

Payload vs. Launch Site



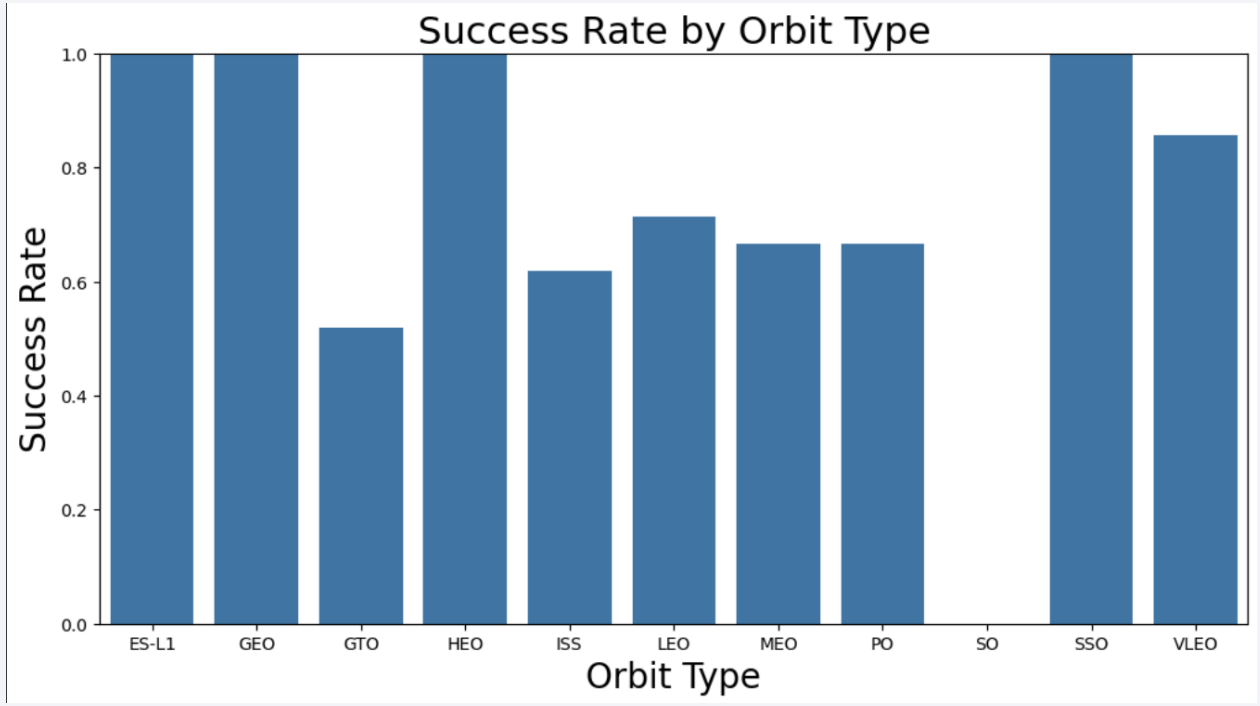
VAFB SLC 4E (Vandenberg Space Force Base, Space Launch Complex 4E): this launch site primarily handles payloads with masses generally below 10,000 kg. There are indeed no rockets launched for heavy payload mass (greater than 10,000 kg) from this site. This might suggest a focus on specific types of missions, possibly polar orbits, which often involve lighter satellites.

CCAFS SLC 40 (Cape Canaveral Space Force Station, Space Launch Complex 40): This site handles a wide range of payload masses, including many heavy payloads. It shows a good mix of both successful (Class 1) and unsuccessful (Class 0) landings across various payload masses, but with a general trend towards higher success rates for heavier payloads in later flights.

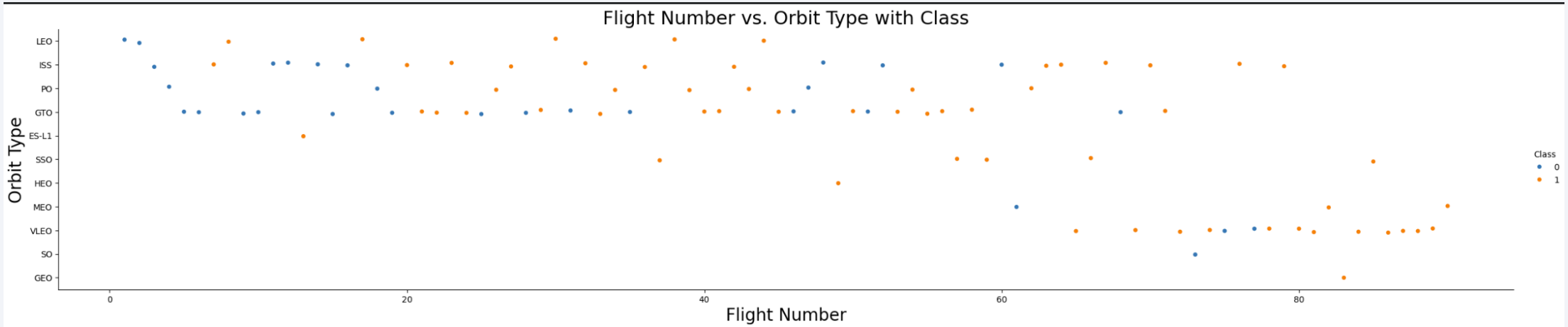
KSC LC 39A (Kennedy Space Center, Launch Complex 39A): This site also accommodates a broad spectrum of payload masses, including very heavy ones. Notably, KSC LC 39A demonstrates a very high success rate for almost all payload masses, indicating its capability and reliability for diverse and often heavier missions.

Overall, the plot indicates that launch sites have different operational envelopes in terms of payload mass. While some sites like CCAFS and KSC handle a full range, VAFB-SLC appears to specialize in lighter payloads. The success rate (Class) also varies by launch site and payload mass, with heavier payloads generally showing better success in later flights at the more active sites.

Success Rate vs. Orbit Type



Flight Number vs. Orbit Type



LEO (Low Earth Orbit): In the early flight numbers, success in LEO missions appears to be mixed. However, as the flight number increases, there's a strong positive correlation with success. The later flights to LEO are almost exclusively successful (Class 1), indicating an improvement in reliability over time for this orbit.

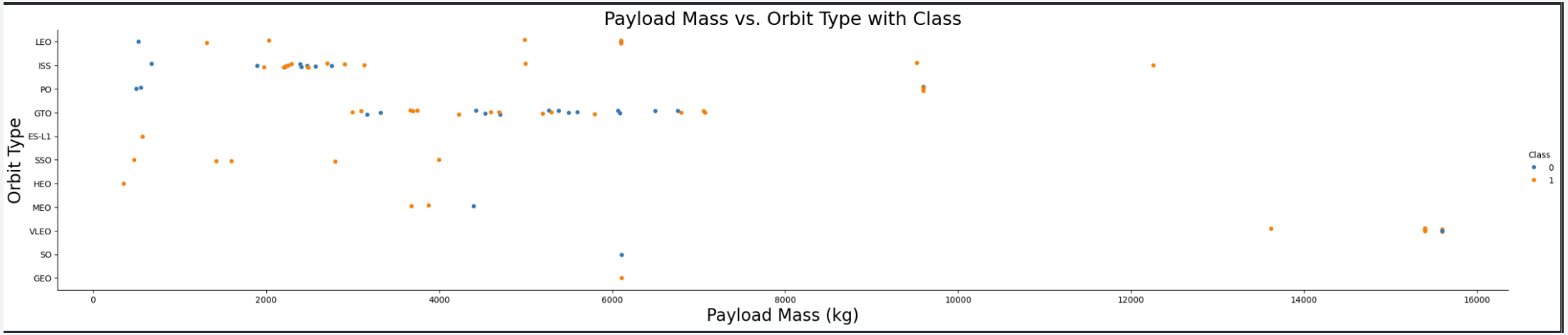
GTO (Geosynchronous Transfer Orbit): For GTO missions, the plot shows a distribution of both successful (Class 1) and unsuccessful (Class 0) landings across various flight numbers. There isn't a clear trend indicating that an increasing flight number directly leads to a higher success rate for GTO, as both outcomes are observed throughout the flight history for this orbit.

ISS (International Space Station): Missions to the ISS generally show a very high success rate (Class 1) regardless of the flight number, from their first appearances in the plot. This suggests consistent success for delivering payloads to the ISS.

VLEO (Very Low Earth Orbit), PO (Polar Orbit), SSO (Sun-Synchronous Orbit), MEO (Medium Earth Orbit), GEO (Geosynchronous Orbit), ES-L1 (Earth-Sun L1 Lagrange point), SO (Solar Orbit): For these orbits, while there might be fewer data points, the successful landings (Class 1) tend to become more prevalent in later flight numbers, reflecting the overall increase in Falcon 9's reliability over time.

In summary, the relationship between Flight Number and success varies by orbit. While LEO shows a clear improvement in success with higher flight numbers, GTO does not exhibit such a strong correlation. Orbits like ISS maintain a high success rate consistently.

Payload vs. Orbit Type



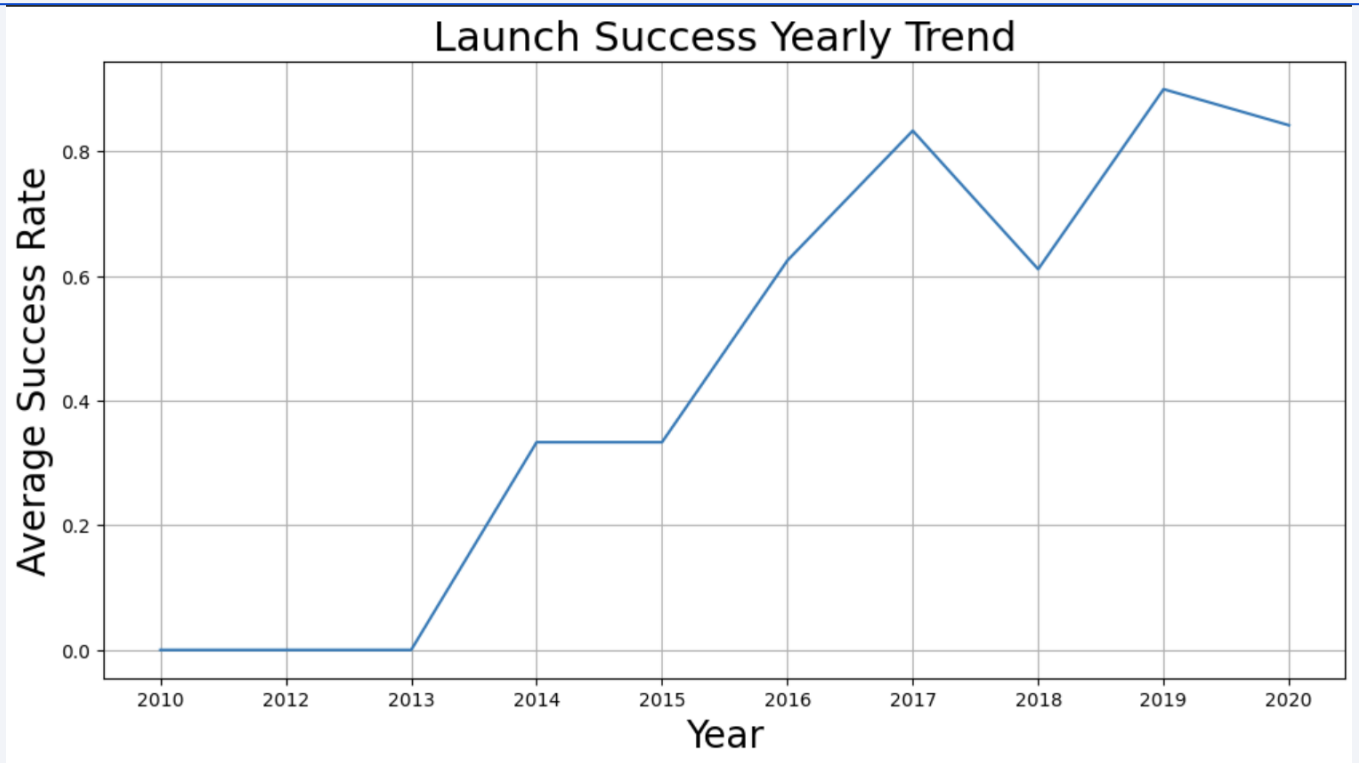
Polar (PO), LEO (Low Earth Orbit), and ISS (International Space Station) orbits with heavy payloads: Your observation is accurate. For these orbits, particularly with heavier payloads, the successful landing rate (Class 1) is indeed higher. This suggests that for missions targeting these orbits, SpaceX has achieved significant reliability even with substantial payload masses. For instance, in LEO and ISS, we see a dense cluster of successful (Class 1, marked in a specific color, likely blue or green) points across a wide range of payload masses, often extending to the heavier end.

GTO (Geosynchronous Transfer Orbit): You correctly noted that for GTO, it's challenging to distinguish a clear pattern between successful and unsuccessful landings based solely on payload mass. The scatter plot for GTO shows a mix of both Class 0 (unsuccessful) and Class 1 (successful) outcomes distributed across various payload masses. This implies that while payload mass is a factor, other variables might be more critical in determining the success of GTO missions, or perhaps the transition to reliable GTO landings took more time or involved different challenges compared to other orbits.

Other Orbits (e.g., SSO, VLEO, MEO): These orbits also show varying trends. Generally, we can observe that as SpaceX gained more experience (which often correlates with higher flight numbers), success rates improved across most orbit types, irrespective of payload mass. However, some orbits inherently pose different challenges or have different mission profiles that influence the payload mass and success outcomes.

In summary, the plot highlights that while heavy payloads can be successfully landed for certain orbits (Polar, LEO, ISS), the GTO orbit presents a more complex picture where payload mass alone doesn't strongly predict landing success.

Launch Success Yearly Trend



The "Launch Success Yearly Trend" line chart clearly illustrates the evolution of Falcon 9's first stage landing success rate over time:

Early Years (2010-2013): In the very early years, the success rate was low, as expected for a new and experimental technology. This period likely involved initial tests and development phases.

Increasing Trend (2013-2020): A significant and consistent upward trend in the average success rate is observed starting from around 2013 and continuing strongly until 2020. This indicates continuous improvements in SpaceX's technology, operational procedures, and experience with first-stage landings.

Plateau/High Success (2020 onwards): By 2020, the success rate appears to have reached a very high level, approaching 1 (or 100%), and seems to plateau there. This suggests that SpaceX has largely perfected the first-stage landing technology, making it a highly reliable and routine operation.

In essence, the plot visually confirms that the Falcon 9 first stage landing capability has matured significantly over the years, transforming from a challenging experimental feat to a highly reliable and successful component of their launch operations.

All Launch Site Names

```
%%sql
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT SUM(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(Payload_Mass__kg_)
45596

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(Payload_Mass__kg_)
2534.6666666666665

First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%%sql
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND Payload_Mass__kg_ > 4000 AND Payload_Mass__kg_ < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%%sql
SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%%sql
SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Payload_Mass__kg_ = (SELECT MAX(Payload_Mass__kg_) FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%%sql
SELECT
  substr(Date, 6, 2) AS Month,
  Booster_Version,
  Launch_Site
FROM SPACEXTABLE
WHERE
  substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Total_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Total_Count DESC;
```

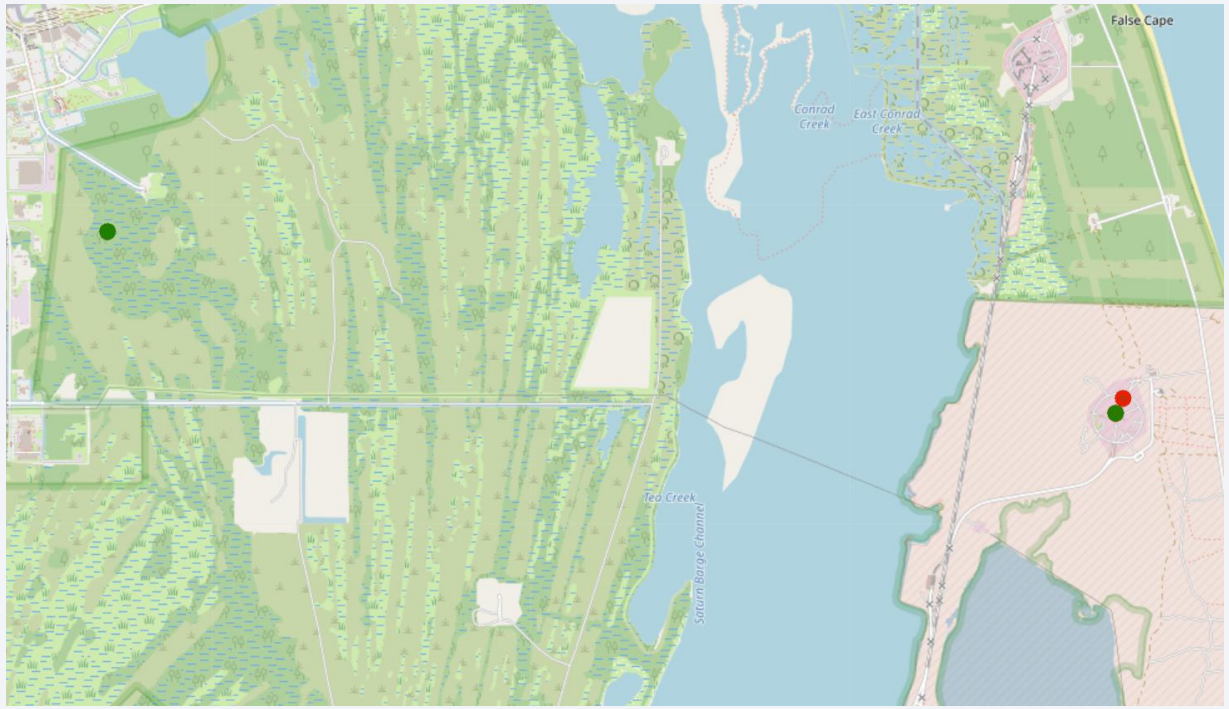
```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

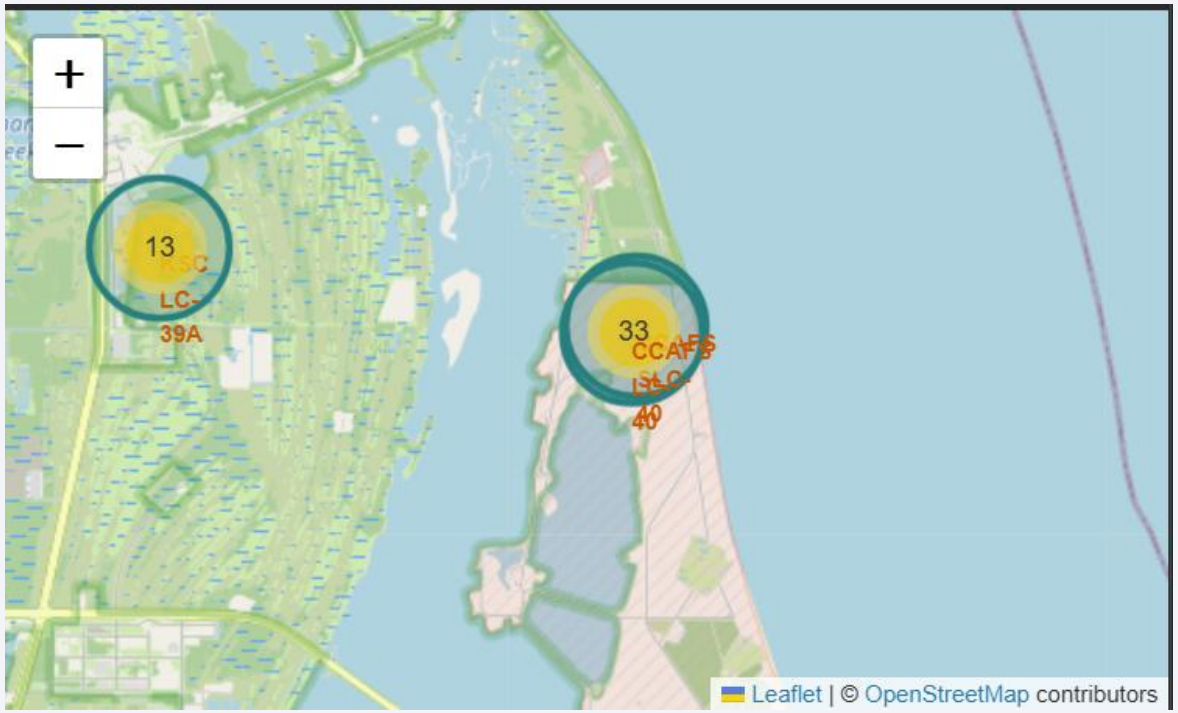
Launch Sites Proximities Analysis

Folium Map Showing all Launch Sites location marker

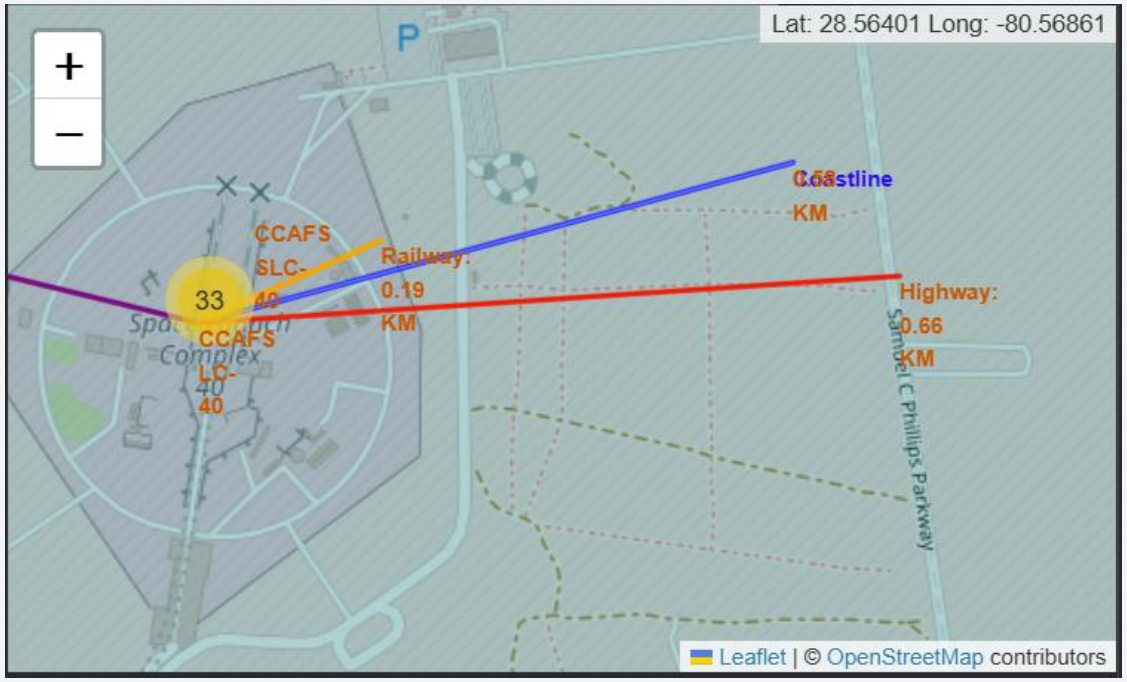


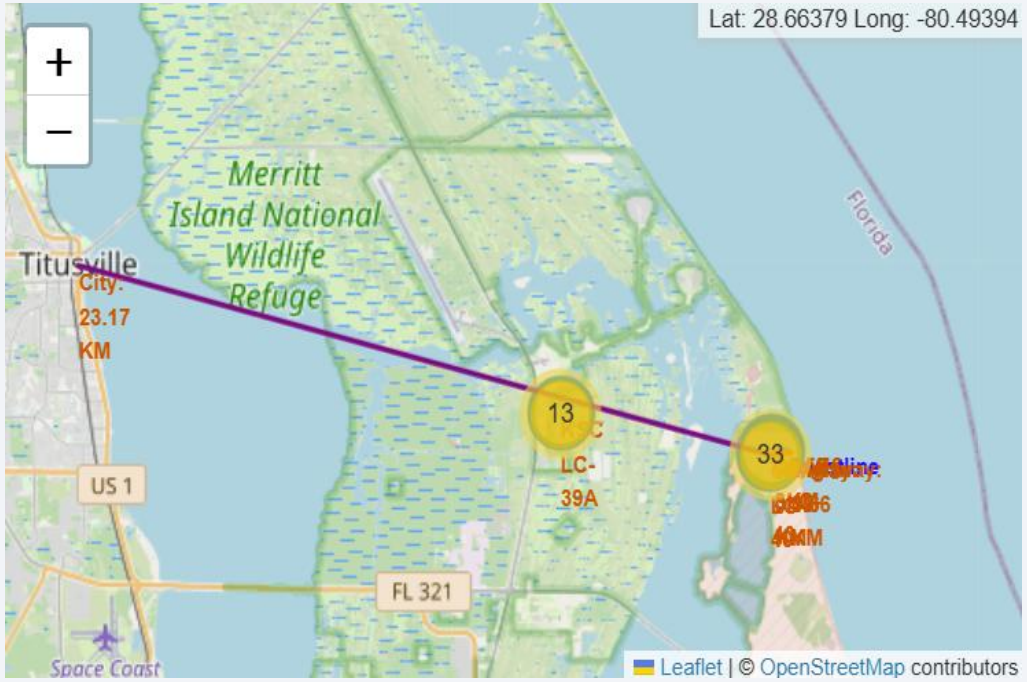
	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0

Folium Map – Coloured Label outcomes of sites



Folium Map showing marker with distance to a closest city, railway, highway,Coast line





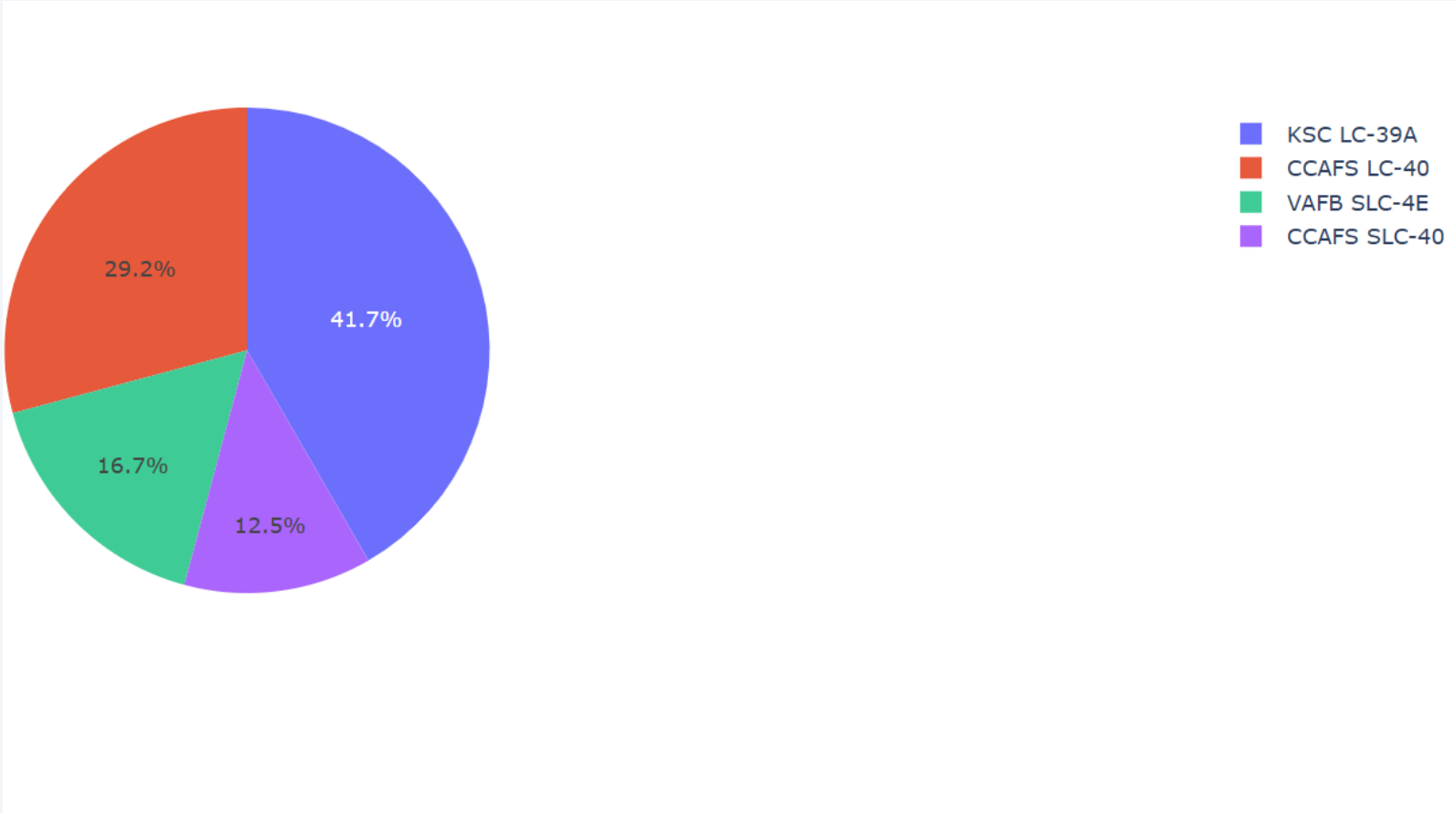
Findings:

Distance from CCAFS LC-40 to City: 23.17 km
Distance from CCAFS LC-40 to Railway: 0.19 km
Distance from CCAFS LC-40 to Highway: 0.66 km
Distance from CCAFS LC-40 to coastline: 0.58 km

Section 4

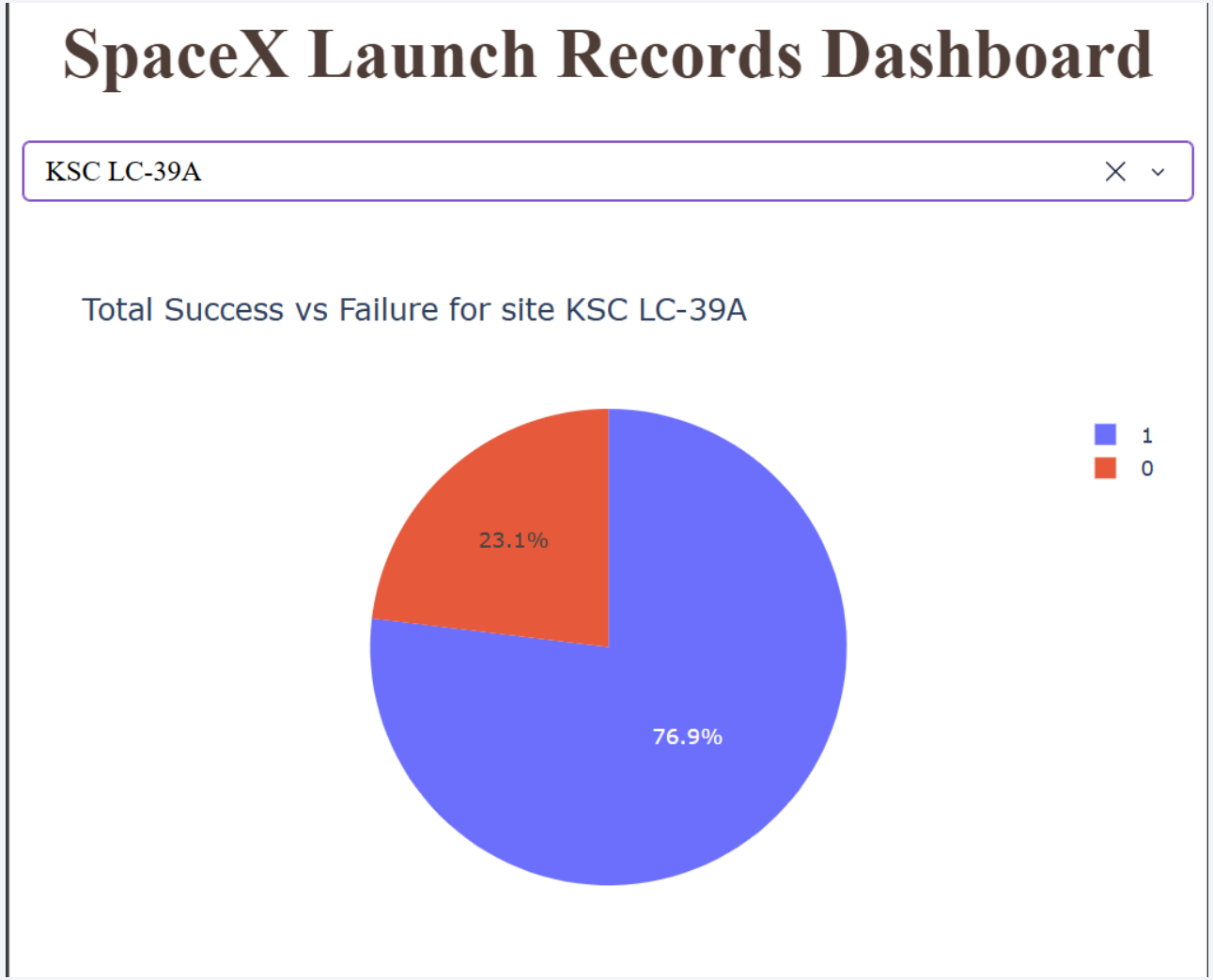
Build a Dashboard with Plotly Dash

Total Success Launches by Site Record Dashboard



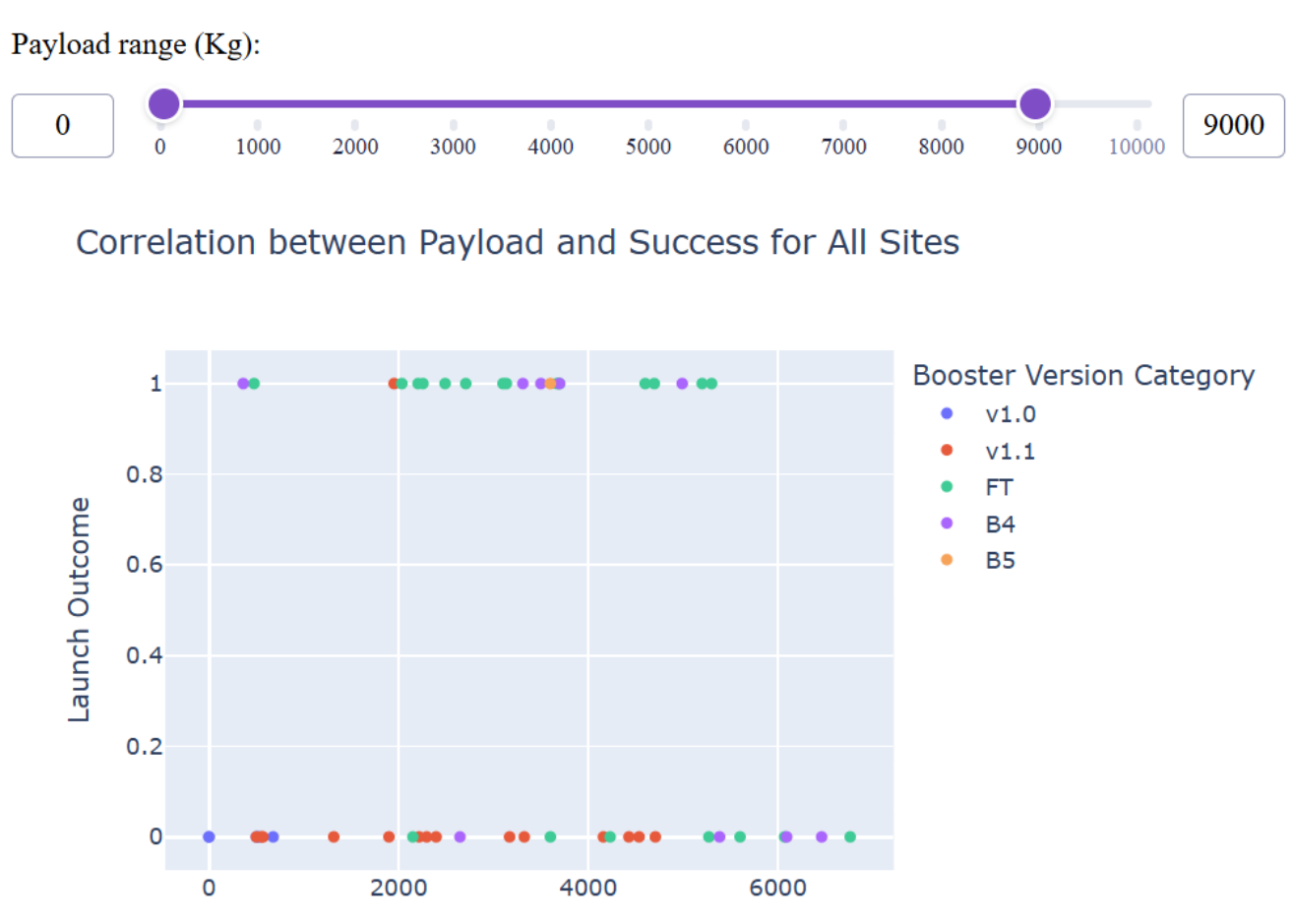
The pie chart emphasizes that **KSC LC-39A is SpaceX's most successful and relied-upon site**.

SpaceX Launch Records Dashboard



- **Operational Insight**: The site's infrastructure supports a high success rate, but the failure percentage highlights the challenges of complex missions often attempted from LC-39A. LC-39A is **reliable with a 3:1 success-to-failure ratio**, but the 23.1% failure share reminds us that it's also the site of some of SpaceX's most ambitious and risk-heavy missions.

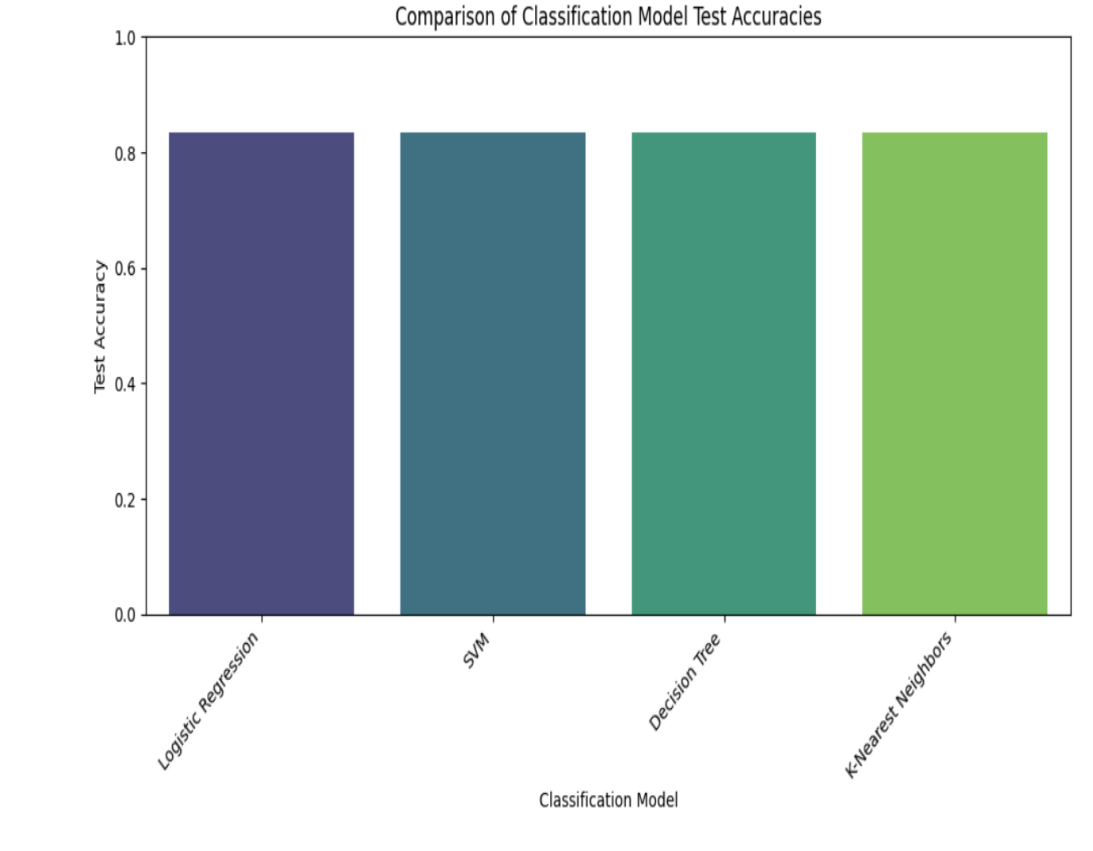
Payload and Success for All sites Dashboard

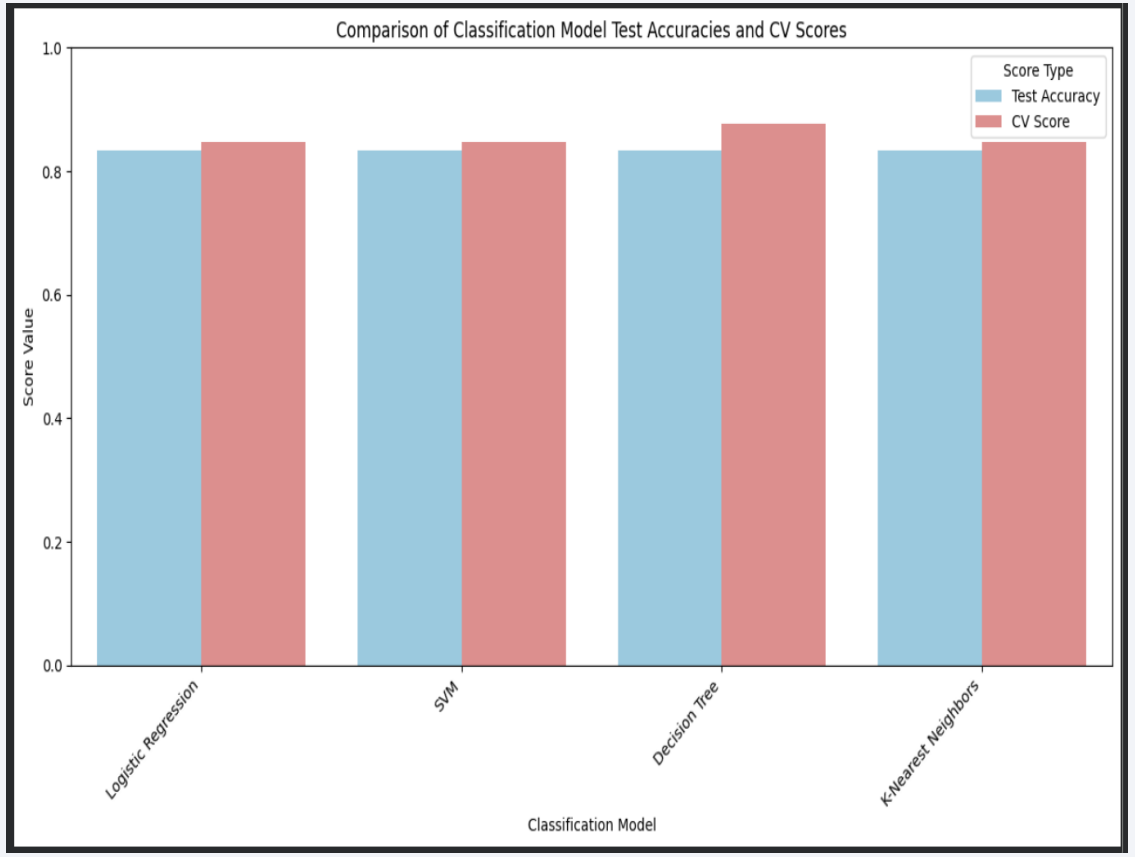


Section 5

Predictive Analysis (Classification)

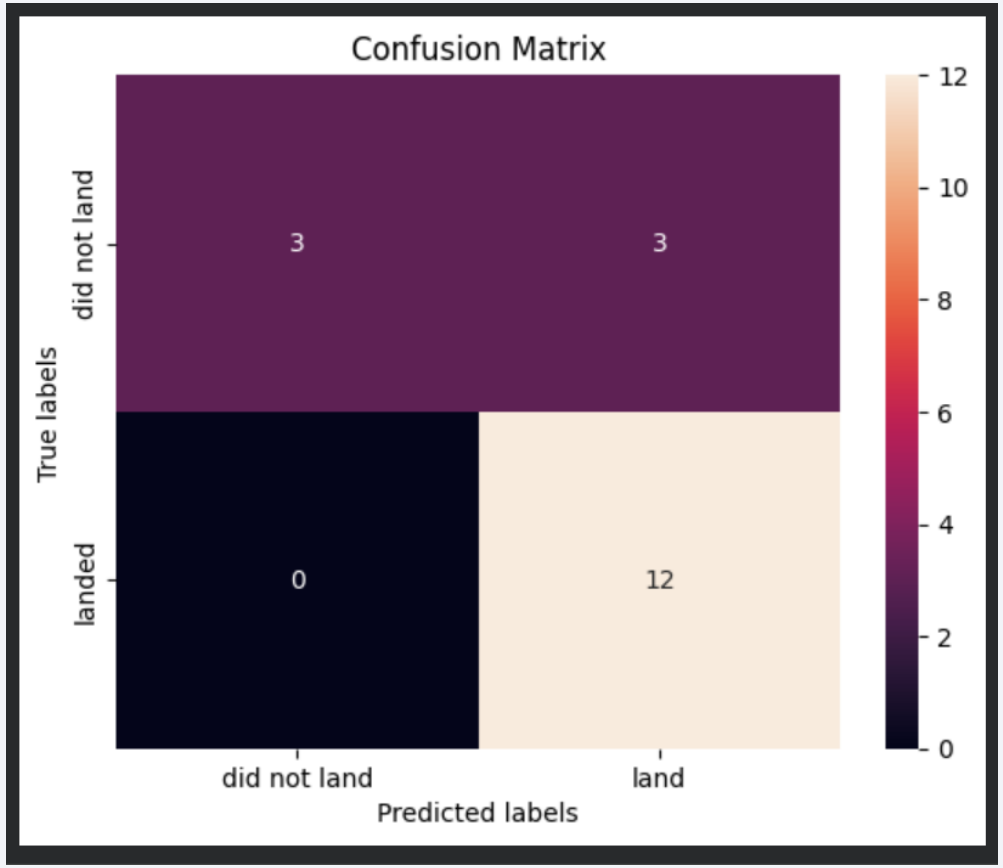
Classification Accuracy





Confusion Matrix

- The **Decision Tree** model is considered the 'best' because the Decision Tree was identified as the top-performing model after a thorough evaluation. We first compared all models based on their **test accuracy**. Since all models achieved the same test accuracy, we then used their **cross-validation (CV) scores** as a tie-breaker. The Decision Tree had the highest CV score (0.8875) among the models with equal test accuracy (0.8333). Therefore, its confusion matrix best represents the performance of the most robust and accurate model found in our analysis.
- **True Negatives (TN) = 3**: The model correctly predicted 3 instances where the rocket **did not land** (Class 0).
- **False Positives (FP) = 3**: The model incorrectly predicted 3 instances where the rocket **landed** (Class 1), but it actually **did not land** (Class 0). This means there were 3 'false alarms' where the model thought a landing would happen, but it didn't.
- **False Negatives (FN) = 0**: The model incorrectly predicted 0 instances where the rocket **did not land** (Class 0), but it actually **landed** (Class 1). This is still excellent, as the model did not miss any actual successful landings.
- **True Positives (TP) = 12**: The model correctly predicted 12 instances where the rocket **landed** (Class 1).
- **Updated Analysis:**
- **High Recall for Landings**: The model still correctly identifies all successful landings (FN=0), which means it has a perfect recall for the 'land' class.
- **Overall Correct Classifications**: The model correctly classified 3 (TN) + 12 (TP) = 15 out of 18 test samples. This gives an accuracy of 15/18 = 0.8333.



Conclusions

1. **Model Performance Summary:**Based on the cross-validation scores, the **Decision Tree** model performed the best with a CV score of **0.8875**. Although all models achieved the same test accuracy of 0.8333 on the given test set, the Decision Tree showed stronger and more consistent performance during cross-validation.

2. The success rate of first-stage landings generally improved with higher FlightNumber, indicating increased reliability over time, even with more massive payloads.
Launch sites like VAFB SLC 4E and KSC LC 39A consistently demonstrated high success rates, especially with increasing flight numbers, while CCAFS SLC 40 also showed improved success in later flights.
VAFB SLC 4E did not conduct launches with payload masses exceeding approximately 10,000 kg, yet maintained a high success rate for its operational range.
Several orbit types, including ES-L1, GEO, HEO, SSO, and VLEO, achieved a 100% success rate, with LEO also showing a very high success rate.
Conversely, SO orbit appeared to have the lowest success rate.
For LEO orbit, success improved with increasing flight numbers, but for GTO orbit, no clear correlation was observed between FlightNumber and landing success.
Heavy payloads in Polar (PO), LEO, and ISS orbits correlated with more frequent successful landings.
There was a consistent upward trend in the overall launch success rate from 2013 through 2020, highlighting continuous improvements in launch and landing capabilities.

3. **Launch Sites**: Four unique launch sites were identified: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40.
Early Missions: The first five records for launch sites starting with 'CCA' were all from CCAFS LC-40, occurring between 2010 and 2013, primarily involving Dragon spacecraft for qualification, demo, or CRS missions targeting LEO or LEO (ISS) orbits. All reported mission outcomes for these were 'Success'.
NASA (CRS) Payload: The total payload mass carried by boosters launched by NASA (CRS) was 45596 kg.
F9 v1.1 Average Payload: The average payload mass for booster version F9 v1.1 was approximately 2534.67 kg.
First Ground Pad Landing: The first successful landing on a ground pad occurred on 2015-12-22.
Successful Drone Ship Landings (4000-6000 kg payload): Specific booster versions F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2 achieved successful drone ship landings within the 4000 kg to 6000 kg payload range.
Mission Outcomes: The majority of missions were 'Success' (98), with isolated instances of 'Failure (in flight)' (1) and other success variants.
Maximum Payload Boosters: A maximum payload mass of 15600 kg was carried by several F9 B5 booster versions, indicating their high capacity.
2015 Landing Failures: No Failure (drone ship) landing outcomes were recorded in the year 2015.
Landing Outcome Trends (2010-2017): During the period from 2010-06-04 to 2017-03-20, 'No attempt' was the most common landing outcome (10 times), followed by 'Success (drone ship)' (8 times) and 'Failure (drone ship)' (5 times).

4. It appears that launch sites are strategically located very close to transportation infrastructure like railways and highways, which would be crucial for transporting rockets, fuel, and other equipment. They are also located very close to the coastline, likely for safety reasons (launches typically occur over water) and to leverage coastal geography. Conversely, launch sites maintain a considerable distance from cities, which is a common practice for safety to mitigate risks associated with launches and potential failures.

Conclusions

- **KSC LC-39A is SpaceX's most successful and relied-upon site**.
- **Operational Insight**: The site's infrastructure supports a high success rate, but the failure percentage highlights the challenges of complex missions often attempted from LC-39A. LC-39A is **reliable with a 3:1 success-to-failure ratio**, but the 23.1% failure share reminds us that it's also the site of some of SpaceX's most ambitious and risk-heavy missions

Appendix

- Python code snippets, SQL queries, charts, Notebook outputs, or data sets have created during this project inserted in previous slides.

Thank you!